

M3: clustering, the symmetry-breaking dichotomy, and the reduction of uniqueness

Abstract

M3 asks for uniqueness of the ground state of $(\mathfrak{A}, \alpha)$. This note delivers three things and is explicit about what it does not deliver.

(1) A correction. Section 4.2 of the strategy note asserted that a uniform gap gives exponential clustering and that “uniqueness of a translation-invariant, exponentially clustering ground state follows from a standard ergodicity argument”. The second half is *false*: clustering gives *extremality*, not uniqueness — two distinct pure phases both cluster. The claim is withdrawn (§3).

(2) A theorem, proved in full. A uniform spectral gap plus the Glimm–Jaffe *exact* light cone gives exponential clustering with an absolute constant and the optimal rate:

$$|\omega_g(AB) - \omega_g(A)\omega_g(B)| \leq 2\|A\| \|B\| e^{-\gamma R/2}, \quad R = \text{dist}(O_1, O_2), \quad \gamma R \geq 1.$$

The proof (§2) is self-contained: an exact spectral-support identity, one explicit test function, and one optimisation. No Lieb–Robinson bound is needed, and no tail is paid, because the light cone is sharp.

(3) A dichotomy. Every limit point of $(\omega_{\tilde{g}_n})$ is $\mathbb{Z}_2$ -invariant [1, Lem. 3.3]. Combined with (2), this forces: *if the $\mathbb{Z}_2$ symmetry is broken, then $\inf_n \text{gap } H(\tilde{g}_n) = 0$* . So the uniform-gap hypothesis (H3) is not a technical convenience — it is equivalent to absence of symmetry breaking along the sequence (§4).

What is not delivered. Uniqueness itself. §5 reduces it to four precisely stated hypotheses (R1)–(R4) and proves the implications between them; (R3), a reverse Osterwalder–Schrader reconstruction, is the one with no proof in the literature. **M3 is therefore half-closed.**

1 Setting

Notation as in [1, 2]. Recall from [1, Fact 1.3] the *exact* light cone $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$, and that $\alpha_t(A) = e^{itH(g)} A e^{-itH(g)}$ for $A \in \mathfrak{A}(O)$ whenever $g \equiv 1$ on $O_{|t|}$. Locality of the net gives $[\mathfrak{A}(O), \mathfrak{A}(O')] = 0$ for disjoint O, O' .

Hypothesis 1.1 (uniform gap). $\gamma := \inf_n \text{gap } H(\tilde{g}_n) > 0$, where $\text{gap } H(g)$ denotes the distance from 0 to the rest of $\text{spec } H(g)$, positive for each g by [4, §2.2–2.3].

2 Exponential clustering from a gap and a sharp light cone

Throughout this section $g \in \mathcal{G}$ is fixed, $\gamma := \text{gap } H(g) > 0$, $P_0 := |\Omega_g\rangle\langle\Omega_g|$, and $A \in \mathfrak{A}(O_1)$, $B \in \mathfrak{A}(O_2)$ with $R := \text{dist}(O_1, O_2) > 0$. We assume $d(g, O_2) > R$, so that the light cone representation of $\alpha_t(B)$ is available for $|t| < R$.

Lemma 2.1 (spectral supports and the locality identity). *Define, for $t \in \mathbb{R}$,*

$$F(t) := \langle A^* \Omega_g, (e^{itH(g)} - P_0) B \Omega_g \rangle, \quad G(t) := \langle B^* \Omega_g, (e^{-itH(g)} - P_0) A \Omega_g \rangle.$$

Then, with $M_0 := \|A\| \|B\|$:

- (i) $F(0) = \omega_g(AB) - \omega_g(A)\omega_g(B)$;
- (ii) $F(t) = \int_{[\gamma, \infty)} e^{itE} d\rho(E)$ and $G(t) = \int_{(-\infty, -\gamma]} e^{itE} d\tilde{\sigma}(E)$ for complex measures with $|\rho|, |\tilde{\sigma}| \leq M_0$;
- (iii) $F(t) = G(t)$ for all $|t| < R$.

Proof. (i) is immediate from P_0 and $\|\Omega_g\| = 1$.

(ii) Put $\rho(\cdot) := \langle A^* \Omega_g, P_{H(g)}(\cdot)(1 - P_0) B \Omega_g \rangle$. Since $\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty)$ and $(1 - P_0)$ removes the eigenvalue 0, $\text{supp } \rho \subseteq [\gamma, \infty)$, and $|\rho|(\mathbb{R}) \leq \|A^* \Omega_g\| \|(1 - P_0) B \Omega_g\| \leq M_0$. The statement for G follows the same way after reflecting $E \mapsto -E$.

(iii) Using $e^{-itH(g)} \Omega_g = \Omega_g$,

$$\langle A^* \Omega_g, e^{itH(g)} B \Omega_g \rangle = \langle \Omega_g, A e^{itH(g)} B e^{-itH(g)} \Omega_g \rangle = \omega_g(A \alpha_t(B)),$$

$$\langle B^* \Omega_g, e^{-itH(g)} A \Omega_g \rangle = \langle \Omega_g, e^{itH(g)} B e^{-itH(g)} A \Omega_g \rangle = \omega_g(\alpha_t(B) A),$$

while both P_0 terms equal $\omega_g(A)\omega_g(B)$. Hence $F(t) - G(t) = \omega_g([A, \alpha_t(B)])$. For $|t| < R$ the light cone gives $\alpha_t(B) \in \mathfrak{A}(O_{2,|t|})$ with $O_{2,|t|} \cap O_1 = \emptyset$, so the commutator vanishes. $\square$

Lemma 2.2 (the test function). *For $\alpha > 0$ and $\varepsilon > 0$ set $f_{\alpha, \varepsilon}(t) := \frac{1}{2\pi i} \frac{e^{-t^2/2\alpha}}{t - i\varepsilon} \in L^1(\mathbb{R})$.*

Then

$$\check{f}_{\alpha, \varepsilon}(E) := \int_{\mathbb{R}} f_{\alpha, \varepsilon}(t) e^{itE} dt = \int_0^\infty \sqrt{\frac{\alpha}{2\pi}} e^{-\alpha(E-E')^2/2} e^{-\varepsilon E'} dE',$$

so that $0 \leq \check{f}_{\alpha, \varepsilon} \leq 1$ and, as $\varepsilon \downarrow 0$, $\check{f}_{\alpha, \varepsilon}(E) \rightarrow \Phi(\sqrt{\alpha}E)$, Φ the standard normal distribution function. Moreover $\int_{|t| \geq R} |f_{\alpha, \varepsilon}| dt \leq \frac{\alpha}{\pi R^2} e^{-R^2/2\alpha}$ uniformly in ε .

Proof. $\int e^{itE} (2\pi i(t - i\varepsilon))^{-1} dt = e^{-\varepsilon E} \mathbf{1}_{E>0}$ by residues, and multiplication by $e^{-t^2/2\alpha}$ becomes convolution with $\sqrt{\alpha/2\pi} e^{-\alpha E^2/2}$; this gives the displayed formula, whence $0 \leq \check{f}_{\alpha, \varepsilon} \leq 1$ and the stated limit by dominated convergence. For the tail, $|t - i\varepsilon| \geq |t|$ gives $\int_{|t| \geq R} |f_{\alpha, \varepsilon}| \leq \frac{1}{2\pi} \int_{|t| \geq R} \frac{e^{-t^2/2\alpha}}{|t|} dt \leq \frac{1}{\pi R} \int_R^\infty e^{-t^2/2\alpha} dt \leq \frac{\alpha}{\pi R^2} e^{-R^2/2\alpha}$, using $\int_R^\infty e^{-t^2/2\alpha} dt \leq \frac{\alpha}{R} e^{-R^2/2\alpha}$. $\square$

Theorem 2.3 (exponential clustering). *With the hypotheses of this section, if $\gamma R \geq 1$ then*

$$|\omega_g(AB) - \omega_g(A)\omega_g(B)| \leq 2 \|A\| \|B\| e^{-\gamma R/2}.$$

Proof. Fix $\alpha, \varepsilon > 0$. Since $f_{\alpha, \varepsilon} \in L^1$ and $|\rho|, |\tilde{\sigma}| < \infty$, Fubini and Lemma 2.1(ii) give

$$\int f_{\alpha, \varepsilon} F = \int_{[\gamma, \infty)} \check{f}_{\alpha, \varepsilon} d\rho, \quad \int f_{\alpha, \varepsilon} G = \int_{(-\infty, -\gamma]} \check{f}_{\alpha, \varepsilon} d\tilde{\sigma}.$$

By Lemma 2.1(iii), $F - G$ vanishes on $(-R, R)$, and $|F|, |G| \leq M_0$, so by Lemma 2.2

$$\left| \int f_{\alpha, \varepsilon} (F - G) \right| = \left| \int_{|t| \geq R} f_{\alpha, \varepsilon} (F - G) \right| \leq 2M_0 \frac{\alpha}{\pi R^2} e^{-R^2/2\alpha}.$$

Letting $\varepsilon \downarrow 0$ and using dominated convergence ($0 \leq \check{f}_{\alpha,\varepsilon} \leq 1$),

$$\left| \int_{[\gamma,\infty)} \Phi(\sqrt{\alpha}E) d\rho(E) \right| \leq \left| \int_{(-\infty,-\gamma]} \Phi(\sqrt{\alpha}E) d\tilde{\sigma}(E) \right| + 2M_0 \frac{\alpha}{\pi R^2} e^{-R^2/2\alpha}. \quad (1)$$

Now use the Gaussian tail bound $\Phi(-x) \leq \frac{1}{2}e^{-x^2/2}$ for $x \geq 0$. On $(-\infty, -\gamma]$ we have $\Phi(\sqrt{\alpha}E) \leq \Phi(-\sqrt{\alpha}\gamma) \leq \frac{1}{2}e^{-\alpha\gamma^2/2}$, so the first term on the right of (1) is at most $\frac{1}{2}M_0e^{-\alpha\gamma^2/2}$. On $[\gamma, \infty)$ we have $|1 - \Phi(\sqrt{\alpha}E)| \leq \frac{1}{2}e^{-\alpha\gamma^2/2}$, so the left side of (1) is at least $|\rho([\gamma, \infty))| - \frac{1}{2}M_0e^{-\alpha\gamma^2/2} = |F(0)| - \frac{1}{2}M_0e^{-\alpha\gamma^2/2}$. Therefore

$$|F(0)| \leq M_0 \left(e^{-\alpha\gamma^2/2} + \frac{2\alpha}{\pi R^2} e^{-R^2/2\alpha} \right).$$

Choosing $\alpha = R/\gamma$ equalises the exponents at $\gamma R/2$ and yields $|F(0)| \leq M_0 e^{-\gamma R/2} (1 + \frac{2}{\pi\gamma R}) \leq 2M_0 e^{-\gamma R/2}$ when $\gamma R \geq 1$. Lemma 2.1(i) finishes the proof. $\square$

Remark 2.4. Nothing in the proof is special to ω_g : replacing $(\Omega_g, H(g))$ by $(\Omega_\omega, H_\omega)$ gives the same bound for any ground state ω of $(\mathfrak{A}, \alpha)$ whose GNS Hamiltonian has a gap γ above 0. Note also where the sharp light cone is used — Lemma 2.1(iii) holds *exactly* for $|t| < R$. With only a Lieb–Robinson bound one would pay an extra tail here, and the rate $\gamma/2$ would degrade to $\gamma/2v$ with additional error terms.

Corollary 2.5. *Under Hypothesis 1.1, the bound of Theorem 2.3 holds for every $\omega_{\tilde{g}_n}$ with the same γ , and hence for every limit point of $(\omega_{\tilde{g}_n})$ in the topology of [1, Cor. 4.3].*

Remark 2.6 (the rate is off by a factor 2, and this is intrinsic to the method). For the free field of mass m one has $\gamma = m$, while the connected two-point function decays as e^{-mR} (the kernel of $(2\mu)^{-1}$ in one dimension behaves like $K_0(mR)$). So the true rate is γ , whereas Theorem 2.3 delivers $\gamma/2$. The loss comes from the optimisation $\alpha = R/\gamma$, which balances a Gaussian energy cutoff against a Gaussian spatial tail and is the same factor-2 loss present in the Nachtergaele–Sims argument. It is harmless for §4, where only $\gamma > 0$ matters, but should be remembered when Theorem 2.3 is fed into a rate computation: combined with M2, whose sharp exponent is $2m_1$, one should not expect the two constants to compose naively.

3 Clustering does not give uniqueness

Remark 3.1 (withdrawal of a claim). The strategy note, §4.2, argued: uniform gap $\Rightarrow$ exponential clustering $\Rightarrow$ uniqueness “by a standard ergodicity argument”. The first implication is Theorem 2.3. The second is false. What ergodicity gives is that a translation-invariant clustering state is *extremal* in the set of translation-invariant states [3, Thm. 4.3.17]. Extremality is compatible with non-uniqueness: in a phase with broken $\mathbb{Z}_2$ symmetry there are two extremal ground states $\omega_\pm$, and *both* cluster exponentially. Clustering separates pure states from mixtures; it does not count the pure states.

What clustering does give is the following, which is genuinely useful because the limit points produced by [1, 2] are $\mathbb{Z}_2$ -invariant.

4 The symmetry-breaking dichotomy

Hypothesis 4.1 (structure of the broken phase). P is even, and there exist ground states $\omega_{\pm}$ of $(\mathfrak{A}, \alpha)$ with $\theta\omega_{\pm} = \omega_{\mp}$, both translation invariant and clustering, and an $A = A^* \in \mathfrak{A}(O)$ with $\omega_{\pm}(A) = \pm m$, $m \neq 0$; moreover the $\mathbb{Z}_2$ -invariant ground states are exactly the segment $\{\lambda\omega_+ + (1 - \lambda)\omega_-\}$ with $\lambda = \frac{1}{2}$.

Hypothesis 4.1 is what [6] is designed to establish at strong coupling.

Theorem 4.2 (dichotomy). *Assume P even and Hypothesis 4.1. Then $\inf_n \text{gap } H(\tilde{g}_n) = 0$; that is, Hypothesis 1.1 fails.*

Proof. Suppose $\gamma := \inf_n \text{gap } H(\tilde{g}_n) > 0$. By [1, Cor. 4.3] and [2, Cor. 4.1] there is a limit point ω of $(\omega_{\tilde{g}_n})$, a ground state of $(\mathfrak{A}, \alpha)$, and by [1, Lem. 3.3] it is $\mathbb{Z}_2$ -invariant. By Corollary 2.5, ω satisfies the bound of Theorem 2.3, so for A as in Hypothesis 4.1,

$$|\omega(A\tau_a(A)) - \omega(A)^2| \leq 2\|A\|^2 e^{-\gamma \text{dist}(O, O+a)/2} \xrightarrow{|a| \rightarrow \infty} 0.$$

By Hypothesis 4.1, $\omega = \frac{1}{2}(\omega_+ + \omega_-)$, so $\omega(A) = 0$ while

$$\omega(A\tau_a(A)) = \frac{1}{2}[\omega_+(A\tau_a(A)) + \omega_-(A\tau_a(A))] \xrightarrow{|a| \rightarrow \infty} \frac{1}{2}[m^2 + m^2] = m^2 \neq 0,$$

using clustering and translation invariance of $\omega_{\pm}$. This contradicts the previous display. $\square$

Remark 4.3. Theorem 4.2 is the expected physics — in a symmetry-broken phase the finite-volume gap closes (tunnelling splitting) — but it is worth having as a theorem, for two reasons. First, it shows Hypothesis 1.1 is not a removable technicality: it is *equivalent* to absence of symmetry breaking along the sequence, so any proof of (H3) must be a proof that the phase is unbroken. Second, it is a nontrivial internal consistency check on the M0–M2 architecture: the $\mathbb{Z}_2$ -invariance of the limit points, which came for free from $\Omega_g > 0$, is exactly what makes the contradiction work.

5 Reduction of uniqueness

Write ω_{∞} for the state obtained from the infinite-volume Euclidean measure ν .

Hypothesis 5.1.

- (R1) (*Euclidean uniqueness.*) The infinite-volume φ_2^4 measure ν is the unique translation-invariant solution of the relevant DLR / stochastic-quantisation characterisation, in the regime considered.
- (R2) (*Global Markov property.*) ν has the global Markov property, equivalently the time-zero fields are cyclic, so that ν determines a state ω_{∞} on the time-zero algebra $\mathfrak{A}$.
- (R3) (*Reverse reconstruction.*) Every locally normal, translation-invariant ground state ω of $(\mathfrak{A}, \alpha)$ has Wightman functions that are tempered and whose analytic continuation yields Schwinger functions of a measure ν_{ω} satisfying the characterisation in (R1).
- (R4) (*Translation invariance of the limit.*) Every limit point of $(\omega_{\tilde{g}_n})$ is translation invariant.

Theorem 5.2. *Assume (R1)–(R4). Then $(\mathfrak{A}, \alpha)$ has a unique locally normal translation-invariant ground state, namely ω_∞ , and $\omega_{\tilde{g}_n} \rightarrow \omega_\infty$ in norm on every $\mathfrak{A}(O)$.*

Proof. Let ω be a locally normal translation-invariant ground state. By (R3) it yields ν_ω satisfying the characterisation of (R1); by (R1), $\nu_\omega = \nu$; by (R2), ν determines a unique state on $\mathfrak{A}$, so $\omega = \omega_\infty$. Thus the ground state is unique. By [1, Cor. 4.3] and [2, Cor. 4.1] every limit point of $(\omega_{\tilde{g}_n})$ is a locally normal ground state, and by (R4) it is translation invariant, hence equals ω_∞ ; [2, Cor. 4.2] upgrades this to convergence of the sequence. $\square$

Remark 5.3 (where the hypotheses come from, and which is open). (R1) is available: Bauerschmidt–Dagallier–Weber (uniqueness of the invariant measure of the φ_2^4 SPDE up to the critical temperature) and Albeverio–Høegh-Krohn–Zegarlinski (Prop. 7.2: uniqueness at differentiability points of the pressure in the external field). (R2) is available at weak coupling: Albeverio–Høegh-Krohn–Zegarlinski prove the global Markov property for weak-coupling φ_2^4 , and by Albeverio–Høegh-Krohn the global Markov property is equivalent to cyclicity of the time-zero fields. (R4) would follow from uniqueness itself by [1, Cor. 2.3], so it is not an independent assumption once (R1)–(R3) give uniqueness within the translation-invariant class; alternatively it can be enforced by averaging, at the cost of [1, Rem. 3.4].

(R3) is the open one. It is a reconstruction in the direction opposite to Osterwalder–Schrader: from an algebraic ground state to a Euclidean measure. Positivity of the energy supplies the analytic continuation; what must be supplied in addition is temperedness (for which [5] Thm A_g is the natural input) and the identification of the resulting Schwinger functions as solutions of the correct Dyson–Schwinger hierarchy, which is where the interaction enters. No proof of (R3) exists in the literature.

6 Status

- **Proved here:** Theorem 2.3 (exponential clustering, absolute constant, rate $\gamma/2$, no Lieb–Robinson input); Theorem 4.2 (uniform gap fails in a symmetry-broken phase).
- **Corrected here:** the uniqueness claim of the strategy note §4.2 is withdrawn (Remark 3.1).
- **Reduced here:** uniqueness follows from (R1)–(R4), Theorem 5.2.
- **Open:** (R3). **M3 is half-closed.** The purely Hamiltonian route of strategy §4.2 is dead as stated; the Euclidean route of §4.1 survives and now has a precise statement of its single missing ingredient.

References

- [1] *Limit points of the finite-volume ground states of $P(\varphi)_2$ are ground states of the quasi-local dynamics*, Milestone M0, ../m0/m0.tex.
- [2] *The local number bound without space-averaging*, Milestone M1, ../m1/m1.tex.
- [3] O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 1*, 2nd ed., Springer, 1987.
- [4] J. Glimm and A. Jaffe, *The $\lambda(\phi^4)_2$ quantum field theory without cutoffs. II*, Ann. of Math. **91** (1970) 362–401.

- [5] J. Glimm and A. Jaffe, *The $\lambda\phi_2^4$ quantum field theory without cutoffs. IV*, J. Math. Phys. **13** (1972) 1568–1584.
- [6] J. Glimm, A. Jaffe and T. Spencer, *A convergent expansion about mean field theory I, II*, Ann. Physics **101** (1976) 610–630, 631–669.